What Has TabPFN's Prior Learned?

Project update · DS-GA 3001: Visualization for Machine Learning · November 3, 2026

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Progress Since the Proposal

- **Foundation (Kiet).** Built the shared repository: a data format for all runs, corrected metrics, nested synthetic datasets, a resumable sweep runner for local and Colab use, and a test suite.
- **Metric test (Kiet).** On surfaces of known shape, the proposal's step score S gives a smooth ramp and a sharp edge the same score (1.00 at 0°, 0.71 at 45°). Our new jump share separates them (0.00 vs. 0.98 at 0°; 0.00 vs. 0.96 at 45°).
- **Classic sweep (Nhan).** All 8,960 synthetic runs are complete: 8 datasets, 4 variants, 4 context sizes, 10 seeds, 7 models. Mean jump share: logistic regression 0.00, GP 0.03, SVM 0.04, random forest 0.71, MLP 0.73, gradient boosting 0.98, k-NN 1.00. Random forest and MLP are close on jump share but differ on axis alignment (0.94 vs. 0.79), so we read the two metrics together. An earlier 84-run rerun at the pilot's size landed within 2.0 accuracy points of the proposal's table.
- **Real-data screen (Nhan).** 33 OpenML datasets screened with classic models only over 10 seeds (4,200 fits; 27 at n=100 and 500, 6 at n=100 only). TabPFN has not seen any of them.
- **Model access (Kiet).** TabPFN v2 and 3.5 both ran on a Colab A100 without errors, taking about 3.2 s and 4.8 s per full 200×200 grid job. This single-problem test gives a rough estimate of 4 GPU hours for the planned runs.
- **Flagged for the metric freeze.** Near-flat surfaces leave shape metrics undefined in some classic runs (248 for axis alignment, 15 for jump share), and SVM probabilities are unreliable at very small n.

Response to Feedback

Feedback	Decision	What changed
1. Step score	Replaced	S is renamed axis alignment; jump share now measures sharpness. Jump share depends on its threshold (random forest: 0.94 at 0.01, 0.36 at 0.1), so we will fix the value before reading sweep results and report sensitivity.
2. Statistical strength	Accepted	10 seeds, with paired differences, 95% CIs, effect sizes and pre-set margins. Inconclusive results are reported as such.
3. Two datasets	Both options	Framed as a case study, with roles assigned using classic models only. Proposed: covertype, run_or_walk, kdd_ipums_la_97_small (tree-favoring) and california, eeg_eye_state, kc1 (smooth-favoring). Smooth-favoring data proved scarce, so we relaxed that side's rule after seeing classic results, before TabPFN, and will disclose this. eeg_eye_state needs a re-screen with time-block splits. Fallback: a smaller case study.
4. Classmate study	Removed	Replaced by 3–5 expert walkthroughs reported qualitatively, plus our worked examples.
5. Visualization claim	Accepted	D3 only; Streamlit removed. Claim: a slider shows one boundary evolving as context grows, and compare mode shows $\lvert pA - pB \rvert$, with a worked example of each. A static page cannot run TabPFN, so the slider uses precomputed nested contexts.
6. Model access	Checked	Go for H5, H6 and RQ3, on the result above. TabPFN runs remain on Kiet's Colab.

Scope and Plan

Completed: the foundation, metric test, classic sweep, dataset screen and model access check above. **Remaining:** TabPFN sweeps, real-data tests, statistical analysis, the D3 interface, walkthroughs and the paper.

Dates	Kiet	Nhan
Nov 4–10	Freeze metrics, hypotheses and margins; run TabPFN synthetic sweep, results unread until the freeze	Finalize datasets and re-screen eeg_eye_state; D3 canvas and controls
Nov 11–17	RQ1 and RQ3 analysis; paper skeleton	Small multiples and metrics panel
Nov 18–24	TabPFN real-data runs; RQ2 analysis	Real-data pipeline and baselines; slider and compare mode
Nov 25–Dec 1	Full paper draft; walkthroughs	Story mode; walkthroughs
Dec 2–8	Freeze results Dec 5; revise paper	Clean-clone reproducibility test; polish interface; revise paper
Dec 9–12	Final fixes; submit by Dec 11	Final fixes; Dec 12 is buffer

If time runs short, we cut in order: ensemble-size ablation, noise-column test, H6, the additional datasets, story mode.

GitHub Repository

https://github.com/KNguyen37/tabpfn-inductive-bias (public; the README lists how to reproduce current results)